

Machine Readable News with SFTP Python Example

- Last update: OCT 2024
- Environment: Windows
- Compiler: Python
- Prerequisite: MRN Archive SFTP service account

Overview

Machine Readable News (MRN) is an advanced service for automating the consumption and systematic analysis of news. It delivers deep historical news archives, ultra-low latency structured news and news analytics directly to your applications. This enables algorithms to exploit the power of news to seize opportunities, capitalize on market inefficiencies, and manage event risk.

The MRN data Real-Time news and News Analytics data are available for consumers via the LSEG Real-Time Platform and SFTP connections. While the Real-Time news connection can be accessed programmatically only (we have covered it via the [Introduction to Machine Readable News \(MRN\) with Enterprise Message API \(EMA\)](#) and [Introduction to Machine Readable News with WebSocket API](#) articles), the SFTP connection can be access with any SFTP applications like the [FileZilla](#), [WinSCP](#), or even Windows native `SFTP` command.

However, some developers might need to access the SFTP remote server and get the file programmatically. This jupyter notebook shows a step-by-step guide to access and get the file from the MRN Remote SFTP site using the [Python](#) programming language with the [Paramiko library](#) on the [JupyterLab notebook](#).

Import Libraries

```
In [1]: import paramiko
        from dotenv import dotenv_values
```

Set Parameters

```
In [2]: config = dotenv_values(".env")
        username = config['MACHINE_ID'] #Or input your Machine-ID Manually
        password = config['PASSWORD'] #Or input your Password Manually
        hostname = 'archive.news.refinitiv.com'
        localfilepath = './download'
```

Establish the SSH Client

```
In [3]: ssh = paramiko.SSHClient()
# Automatically add host keys (not secure for production)
ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())
```

Connect to the Remote Server and Establish the SFTP client

VPN Connection

Please be informed that I needs to connect to the MRN SFTP Remote server via Cisco AnyConnect VPN from the LSEG local machine.

I strongly suggest you contact the MRN Support and your local IT Support team to help you with a connect to MRN SFTP from your company environment.

```
In [4]: # If your environment needs to connect to network via Proxy
# import socks
# sock = socks.socksocket()
# sock.set_proxy(
#     proxy_type=socks.SOCKS5,
#     addr="Your_Proxy_URL",
#     port=Your_proxy_port_number,
#     username="Your Proxy Username",
#     password="Your Proxy Username"
# )
sftp = None
try:
    ssh.connect(hostname=hostname, username=username, password=password)
    # If your environment needs to connect to network via Proxy
    # ssh.connect(hostname=hostname, username=username, password=password, sock=sock)
    sftp = ssh.open_sftp()
except Exception as e:
    print(f'Fail on connection :{e}')
```

Check the current directory

```
In [5]: print(sftp.listdir())
```

```
['mpsyh', 'na-gold']
```

Change Directory to get a file

I am demonstrating with **/mpsyh/MI4/CMPNY_REF/BASIC** folder and get the first file from the Remote server.

```
In [7]: try:
    sftp.chdir('/mpsyh/MI4/CMPNY_REF/BASIC')
    #Lists files on this /mpsyh/MI4/CMPNY_REF/BASIC folder
    print(f'lists files on this /mpsyh/MI4/CMPNY_REF/BASIC folder: {sftp.listdir()}')
except Exception as e:
    print(f'Fail on chaning directory :{e}')
```

lists files on this /mpsyh/MI4/CMPNY_REF/BASIC folder: ['MI.Companies.BASIC.04043.txt', 'MI.Companies.Changes.04043.txt']

```
In [50]: # Get the first file name
fileName = sftp.listdir()[0]
```

Downloading the file

```
In [8]: try:
        sftp.get(remotepath = fileName, localpath=f'{localfilepath}\\{fileName}')
    except Exception as e:
        print(f'Fail on get a file :{e}')
```

Read the file

```
In [9]: try:
        with open(f'{localfilepath}\\{fileName}') as finput:
            data = finput.read(200)
            print(data)
    except Exception as e:
        print(f'The error is:{e}')
```

OrgPermID	Name	TRBC	TRBCPermID	TRBCEconomicSector	Ticker	MIC
RIC	Domicile	ExchCtry	ExchCode	InfoCode	region	stat
us	CurrentInstrumentIssuerPermId					
4294970469	ESPN Inc		5330202011	4294951554	Consumer	cycl

Close the SFTP session

```
In [12]: sftp.close()
```

Close this SSHClient

```
In [13]: ssh.close()
```

References

For further details, please check out the following resources:

- [LSEG Developer Community website](#).
- [Machine Readable News website](#).
- [Paramiko official website](#)
- [Paramiko API document](#)
- [Paramiko SFTP API document](#)
- [Paramiko Client API document](#)
- [Paramiko- How to transfer files with Remote System \(SFTP Servers\) using Python blog post](#).
- [How-to: Python Paramiko blog post](#).
- [Paramiko SFTP: A Guide with Examples website](#).

- [Using Paramiko with SOCKS proxy](#) post on [StackOverflow](#).

In []: